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CSCI596 Assignment 4—Parallel Molecular Dynamics

(Part I—Asynchronous Messages)

- **Submit the modified source code, with your modifications clearly marked.**

Both pmd_irecv.c and pmd_irecv.h files are attached as part of this submission.

- Which program runs faster? Repeat the comparison three times and report the average runtime of both programs. **Submit the timing data.**

The asynchronous version is faster than the synchronous version by ~10.5%. Averaged over three runs, the asynchronous version took 0.4997s and the synchronous version 0.5584s (standard deviations 0.0014s and 0.0110s). The timing data pmd_irecv.out is attached as part of this submission and its contents are pasted below:

***** Asynchronous *****

al = 5.129928e+00 5.129928e+00 5.129928e+00
lc = 2 2 2
rc = 2.564964e+00 2.564964e+00 2.564964e+00
nglob = 1728
CPU & COMT = 5.005926e-01 1.278336e-01

***** Synchronous *****

al = 5.129928e+00 5.129928e+00 5.129928e+00
lc = 2 2 2
rc = 2.564964e+00 2.564964e+00 2.564964e+00
nglob = 1728
CPU & COMT = 5.529112e-01 1.656170e-01

***** Asynchronous *****

al = 5.129928e+00 5.129928e+00 5.129928e+00
lc = 2 2 2
rc = 2.564964e+00 2.564964e+00 2.564964e+00
nglob = 1728
CPU & COMT = 4.981113e-01 1.237926e-01

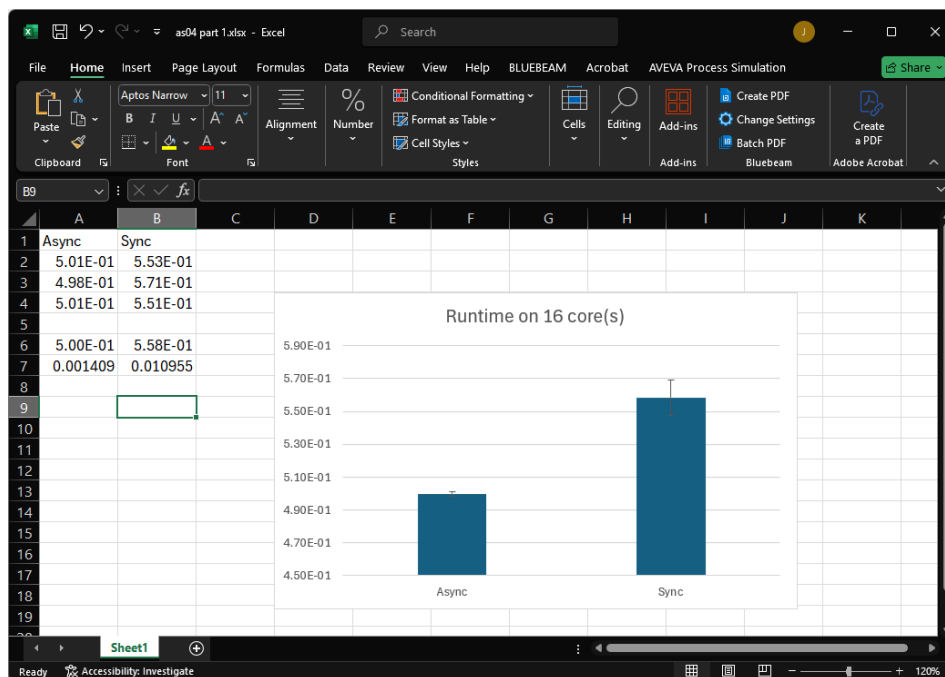
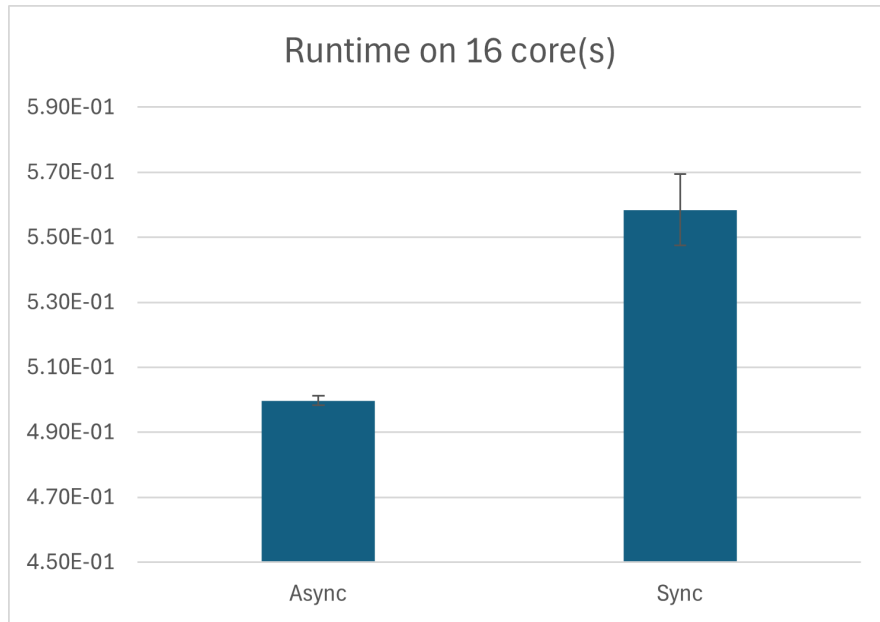
***** Synchronous *****

al = 5.129928e+00 5.129928e+00 5.129928e+00
lc = 2 2 2
rc = 2.564964e+00 2.564964e+00 2.564964e+00
nglob = 1728
CPU & COMT = 5.710159e-01 1.669194e-01

***** Asynchronous *****

al = 5.129928e+00 5.129928e+00 5.129928e+00
lc = 2 2 2

```
rc = 2.564964e+00 2.564964e+00 2.564964e+00
nglob = 1728
CPU & COMT = 5.005084e-01 1.263087e-01
***** Synchronous *****
al = 5.129928e+00 5.129928e+00 5.129928e+00
lc = 2 2 2
rc = 2.564964e+00 2.564964e+00 2.564964e+00
nglob = 1728
CPU & COMT = 5.512776e-01 1.661301e-01
```



(Part II—Communicators)

- **Submit the modified source code (name it `pmd_split.c`), with your modifications clearly marked.**

Both `pmd_split.c` and `pmd_split.h` files are attached as part of this submission.

- **Submit the plot of calculated PDFs at time steps 10, 20, and 30.**

